SEMESTER 2 • ENGINEERING SCIENCE

Basic Electronics

A complete diploma handbook covering semiconductor theory, PN junction diodes, BJT, JFET, MOSFET, UJT, rectifiers, filters, RC wave shaping, clippers, clampers and voltage multipliers.

COURSE CODE
2041

PROGRAMME
Electronics / ECE / Biomedical

SEMESTER
2

CREDITS
3

CATEGORY
Engineering Science

PERIODS / WEEK
3 (L:3,T:0,P:0)

PERIODS / SEMESTER
45

COURSE TITLE
Basic Electronics

45

Periods from semiconductor material to practical circuits

Connect device construction, biasing, carrier motion, characteristic curves, circuit operation and waveform output.

PN Diode BJT JFET MOSFET UJT Wave Shaping

COURSE OVERVIEW

Why Basic Electronics matters

Device physics

Understand energy bands, carriers, doping, depletion region and biasing.

> Carrier movement □□□□□□□□□□□□□□□ device working □□□□□□□□□□□.

Signal control

Use transistors for amplification, switching, voltage control and triggering.

Practical circuits

Build rectifier, filter, differentiator, integrator, clipper, clamper and multiplier concepts.

Electronics in practice: The same concepts appear in regulated power supplies, small-signal amplifiers, sensor interfaces, communication receivers, medical instrumentation front ends, triggering circuits and industrial control panels.

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COURSE DETAILS

Objectives, prerequisites, outcomes and CO-PO mapping

Objectives

1. Introduce semiconductor diodes and application circuits.
2. Illustrate working of different transistors.
3. Introduce linear and nonlinear wave-shaping circuits.

Prerequisites

- Electric circuits and passive components — Fundamentals of Electrical and Electronics, Semester 2.
- Mathematics I & II — Semesters 1 & 2.
- Applied Physics I & II — Semesters 1 & 2.

CO	Description	Hours	Level
CO1	Explain semiconductor theory and semiconductor diodes.	10	Understanding
CO2	Explain BJT and transistor amplifier.	11	Understanding
CO3	Summarize JFET, MOSFET and UJT.	11	Understanding
CO4	Develop linear and nonlinear wave-shaping circuits.	11	Applying
Tests	Series Tests I & II	2	-

CO-PO Mapping

CO	PO1	PO3	Mapping
CO1	2	-	Moderate
CO2	2	-	Moderate
CO3	2	-	Moderate
CO4	3	2	Strong PO1, Moderate PO3

Scale: 3 Strong, 2 Moderate, 1 Weak.

LEARNING ROADMAP

45-period plan

Module	Hours	Main learning
1. Semiconductor and PN diode	10	Bands, carriers, doping, junction, biasing, V-I curve and specifications.
2. BJT and amplifier	11 + Test I	NPN/PNP, modes, gains, configurations, characteristics and amplification.
3. JFET, MOSFET and UJT	11	Voltage control, channel behavior and negative resistance.
4. Rectifiers and wave shaping	11 + Test II	Rectifiers, filters, RC circuits, clipping, clamping and multiplication.

MODULE 1 • CO1

Semiconductor Theory and PN Junction Diode

Module Outcomes

- Describe intrinsic/extrinsic semiconductors.
- Explain junction formation and barrier potential.
- Illustrate bias and V-I characteristics.
- List diode specifications.

Energy-band foundation

Valence electrons occupy the valence band. Electrons in the conduction band can move through the material. The forbidden gap separates them.

> Small band gap □□□□□□□□□□ semiconductor conductivity control □□□□□□□□.

Material	Band condition	Behavior
Conductor	Bands overlap	High conductivity
Semiconductor	Small forbidden gap	Controllable conductivity
Insulator	Large forbidden gap	Very low conductivity

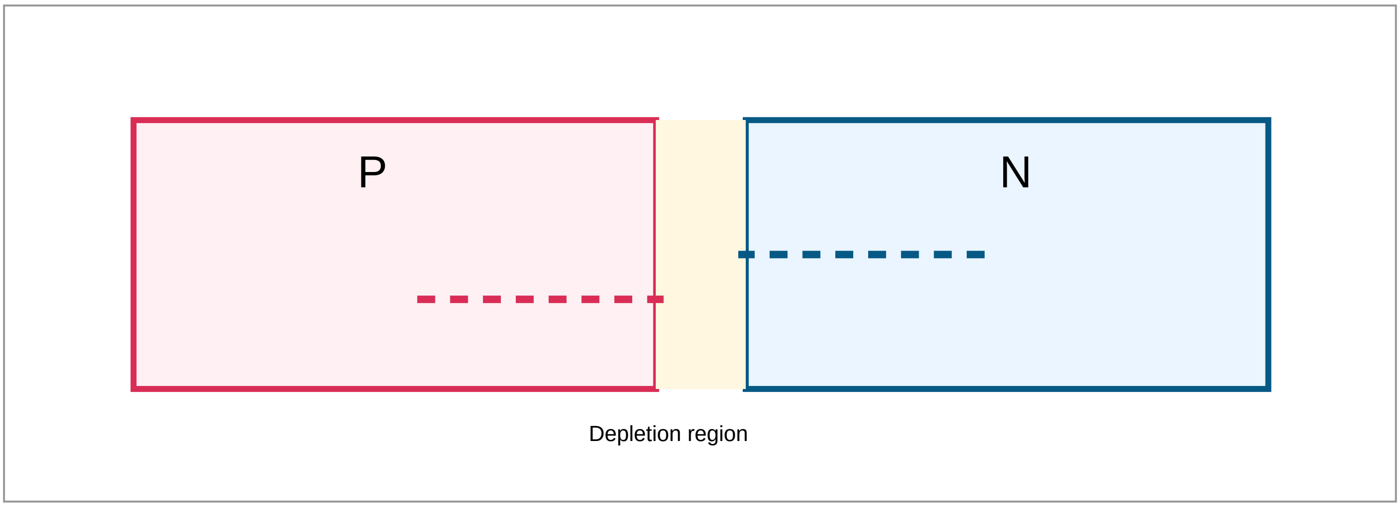
Intrinsic, thermal generation and doping

A pure semiconductor is intrinsic. Thermal energy breaks covalent bonds and creates electron-hole pairs; recombination restores bonds. Pentavalent donor doping creates N-type material with electrons as majority carriers. Trivalent acceptor doping creates P-type material with holes as majority carriers. Both materials remain electrically neutral.

PN-junction formation

Electrons diffuse N→P and holes P→N. Recombination leaves fixed ions and creates a depletion region. The internal electric field produces barrier potential and drift current opposing diffusion.

> Junction □□□□□□□□□ mobile carriers □□□□□□□□□□□□□ depletion region □□□□□□□□□.



Forward and reverse bias

Forward bias

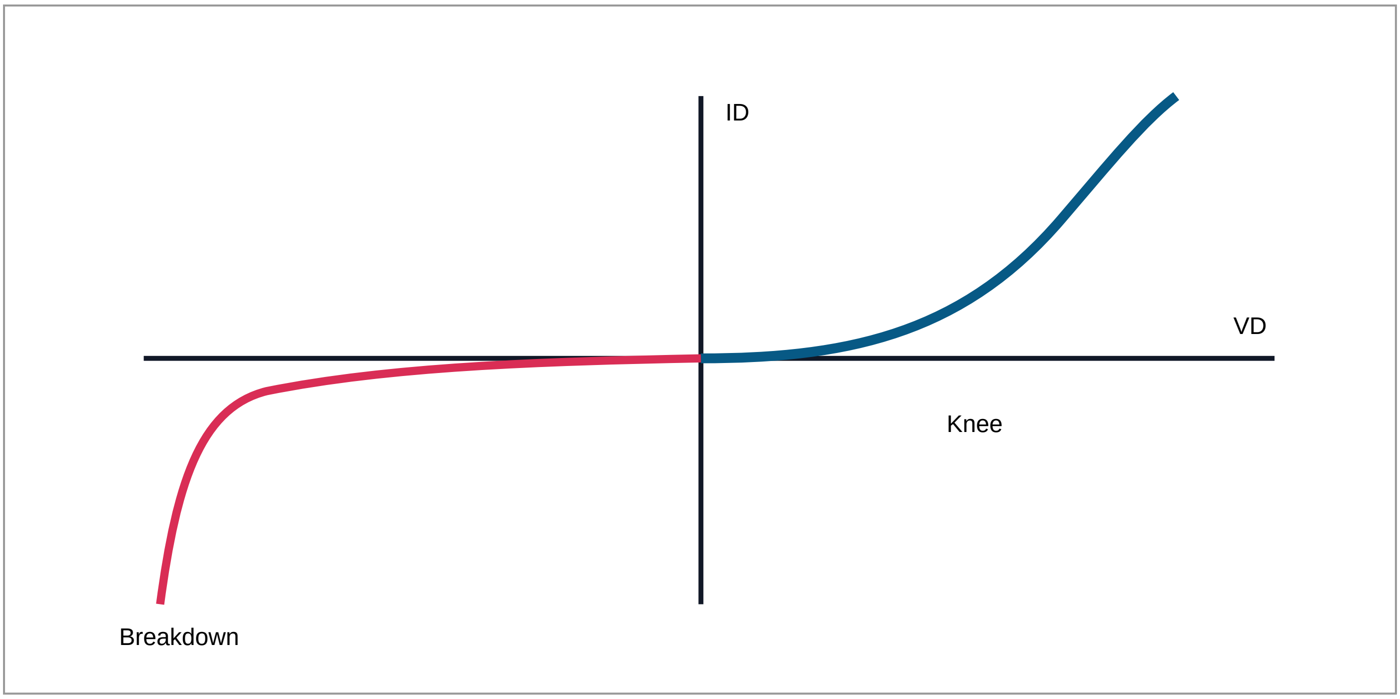
P to positive and N to negative. Depletion width and barrier decrease; current rises rapidly after knee voltage. Typical barrier potential is approximately 0.7 V for silicon and 0.3 V for germanium at room temperature.

Reverse bias

P to negative and N to positive. Depletion width increases; only small reverse saturation current flows until breakdown.

Useful Extra Note — Shockley diode equation: For deeper understanding, diode current can be represented by $I = I_S(e^{qV_D/kT} - 1)$. At room temperature, kT/q ≈ 26 mV. For this diploma syllabus, use it as a concept note; numericals usually use static/dynamic resistance and diode power.

V-I characteristic



Knee voltage marks rapid forward-current rise. Reverse saturation current is minority-carrier leakage. Zener breakdown occurs in heavily doped junctions at lower reverse voltage; avalanche generally occurs at higher voltage.

$$RDC=VD/ID$$

$$rac=\Delta VD/\Delta ID$$

$$PD=VD\times ID$$

Static resistance

VD=0.70 V, ID=10 mA=0.010 A.
RDC=0.70/0.010=70 Ω.

Dynamic resistance

Voltage changes 0.68→0.72 V; current 8→12 mA.
ΔV=0.04 V; ΔI=0.004 A.
rac=0.04/0.004=10 Ω.

Diode specifications

Specification	Meaning

Forward voltage	Voltage drop while conducting.
Maximum forward current	Largest safe current.
Power rating	Maximum safe dissipation.
PIV	Maximum reverse voltage.
Reverse saturation current	Leakage before breakdown.

Expected: energy bands, N/P comparison, junction formation, biasing, V-I curve, RDC/rac/power and rating selection.

MODULE 2 • CO2

Bipolar Junction Transistor and Transistor Amplifier

Outcomes

- Describe BJT structure.
- Explain working and modes.
- Summarize configurations and gains.
- Illustrate characteristics and amplification.

BJT construction

Emitter is heavily doped, base is thin/lightly doped, collector is moderately doped and physically larger. NPN conduction is mainly by electrons; PNP mainly by holes.

> Thin base □□□□□ carriers-□□□□ □□□□□□□□□□ collector-□ □□□□□□.

Mode	EB junction	CB junction	Use
Cut-off	Not forward	Reverse	Switch OFF
Active	Forward	Reverse	Amplifier
Saturation	Forward	Forward	Switch ON

NPN operation

Forward-biased emitter-base junction injects electrons into the thin base. A small fraction recombines, causing base current. Most are swept across the reverse-biased collector-base junction and form collector current.

$$IE=IC+IB$$

$$\alpha=IC/IE \quad \beta=IC/IB \quad \gamma=IE/IB$$

$$\beta=\alpha/(1-\alpha) \quad \alpha=\beta/(\beta+1) \quad \gamma=\beta+1$$

Current calculation

IC=4.8 mA, IB=0.2 mA.
IE=IC+IB=5.0 mA.

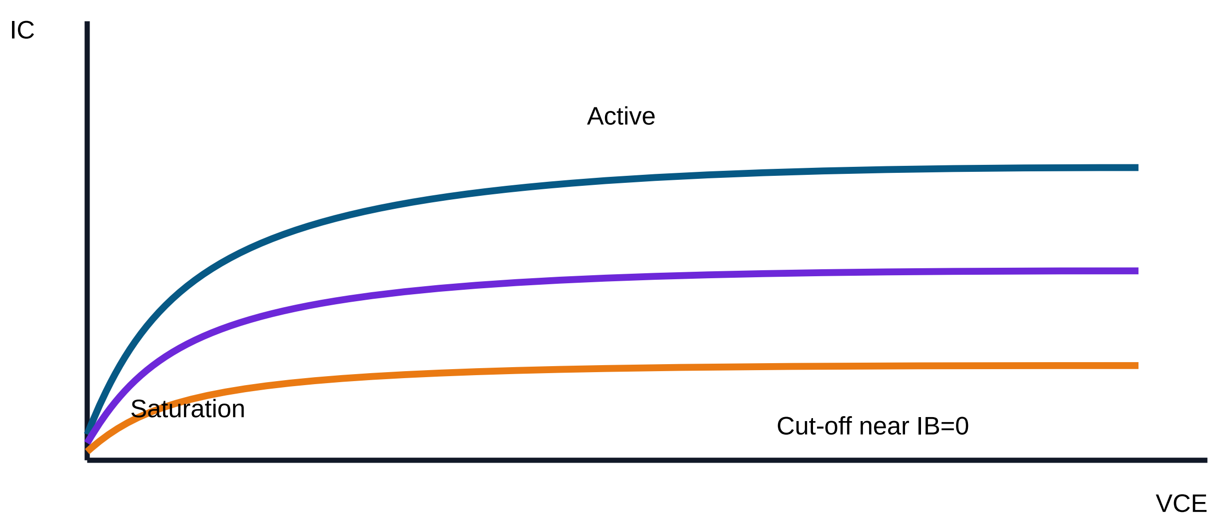
Beta

IC=6 mA, IB=50 μA=0.05 mA.
β=6/0.05=120.

Configurations

Feature	CB	CE	CC
Current gain	α<1	β high	γ very high
Voltage gain	High	High	≈1
Phase reversal	No	180°	No
Use	High frequency	General amplifier	Buffer

CE characteristics and amplifier



CE is important because it provides useful current and voltage gain, giving high power gain. In active region a small base-signal variation produces a larger, inverted collector-voltage output.

Expected: NPN/PNP structure, modes, IE relation, α/β conversion, CB-CE-CC comparison, CE curves and transistor amplifier.

MODULE 3 • CO3

JFET, MOSFET and UJT

Outcomes

- Describe structures.
- Explain operation.
- Illustrate characteristics.
- Compare BJT and FET.

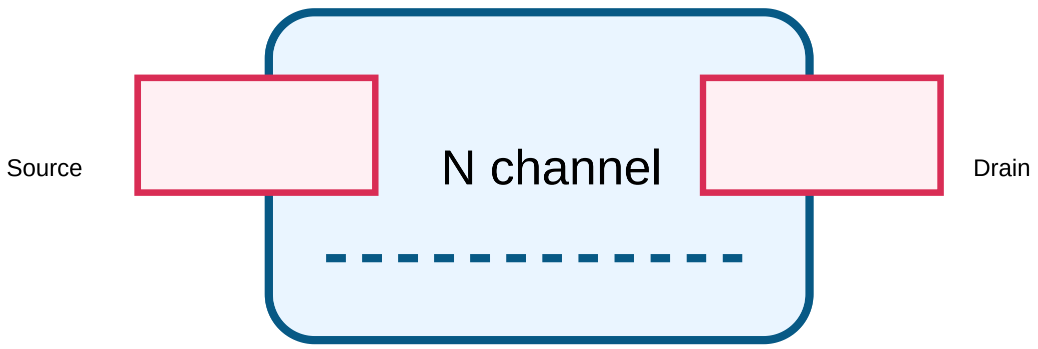
Field-effect concept

FETs are voltage-controlled and mainly unipolar. Gate current is extremely small, producing high input impedance.

> Gate voltage channel current control □□□□□□□□□□.

N-channel JFET

The reverse-biased P gate controls an N channel between source and drain. Increasing negative VGS widens depletion regions and reduces ID. In the ohmic region current rises with VDS; after pinch-off it is nearly constant; at VGS(off) the channel cuts off.



Useful Extra Note — JFET transfer relation: A common introductory relation for an N-channel JFET is $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$. It helps interpret the transfer characteristic; avoid overusing it when the question asks only qualitative working.

JFET transfer characteristic

Given: $I_{DSS}=10$ mA, $V_P=-4$ V, $V_{GS}=-2$ V.
Formula: $I_D=I_{DSS}(1-V_{GS}/V_P)^2$.
$I_D=10(1-(-2/-4))^2=10(0.5)^2=2.5$ mA.

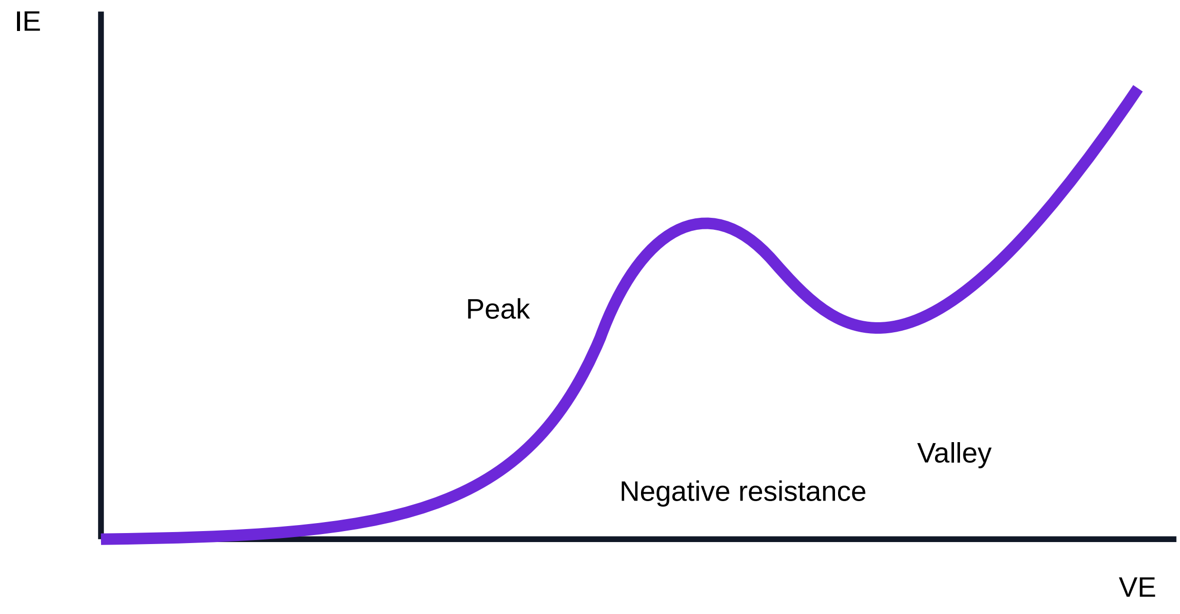
Depletion MOSFET

An insulated gate controls an existing N channel. At $V_{GS}=0$ it conducts; negative V_{GS} depletes and reduces current; positive V_{GS} enhances current. The insulated gate gives very high input impedance.

UJT

A UJT has emitter E and bases B1/B2. Its N bar behaves as R_{B1} and R_{B2}. When emitter voltage reaches the peak point, carrier injection reduces R_{B1} and creates a negative-resistance region until the valley point.

$$\eta=R_{B1}/(R_{B1}+R_{B2})$$



Intrinsic stand-off ratio

$R_{B1}=6$ kΩ, $R_{B2}=4$ kΩ.
$\eta=6/(6+4)=0.6$.

Device-selection shortcut: Use a BJT for current gain and simple switching, a JFET for low-noise high-input-impedance signal stages, a MOSFET for extremely high input impedance or fast voltage-controlled switching, and a UJT for triggering or relaxation-oscillator applications.

Feature	BJT	JFET	MOSFET	UJT
Control	Base current	Gate voltage	Insulated gate	Emitter trigger
Input impedance	Lower	High	Very high	Special-purpose
Main use	Amplifier/switch	High-Z amplifier	Voltage control	Trigger/oscillator

Expected: JFET structure, pinch-off and curves; depletion MOSFET; UJT equivalent circuit, η, peak/valley; BJT-FET comparison.

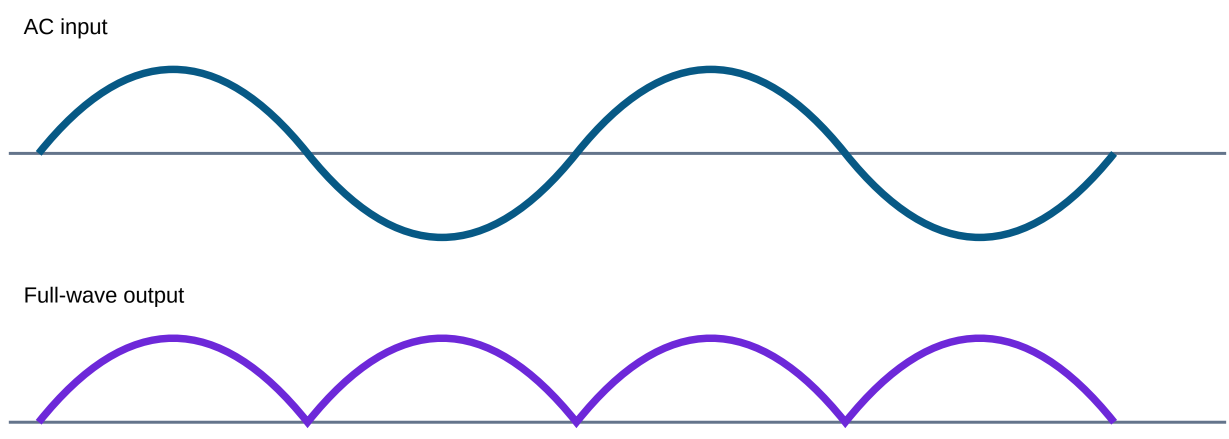
MODULE 4 • CO4

Rectifiers, Filters and Wave-Shaping Circuits

Rectifier comparison

Standard values below assume ideal diodes and resistive load. Vm is peak secondary voltage.

Feature	Half-wave	Centre-tapped	Bridge
VDC	Vm/π	2Vm/π	2Vm/π
Ripple	1.21	0.482	0.482
PIV/diode	Vm	2Vm	Vm
TUF	≈0.287	≈0.693	≈0.812
Maximum efficiency	≈40.6%	≈81.2%	≈81.2%
Transformer	Simple	Centre tap	No centre tap



Filters

Capacitor

Charges at peaks and discharges through load, filling valleys.

Inductor

Series inductance opposes ripple current.

π filter

C-L-C combination gives strong ripple reduction.

RC differentiator and integrator

$$\tau = RC$$

$$\text{Differentiator: } RC \ll T \quad \text{Integrator: } RC \gg T$$

Differentiator

Series C, shunt R, output across R. Square-wave edges produce positive and negative spikes.

Integrator

Series R, shunt C, output across C. Square input produces approximate triangular output.

Time constant

$R = 10\ k\Omega$, $C = 0.01\ \mu F$.
$\tau = 10{,}000 \times 0.01 \times 10^{-6} = 0.1$ ms.
For $T = 10$ ms, $RC \ll T$, so differentiation is possible.

Clippers and clampers

A shunt clipper diverts an unwanted positive or negative portion. A bias source sets clipping level; a combination clipper limits both sides; a slicer passes a selected window. A clamper shifts the whole waveform DC level using diode, capacitor and resistor.

Feature	Clipper	Clamper
Action	Removes waveform portion	Shifts DC level

Peak-to-peak	Usually changes	Ideally unchanged
Key component	Diode + resistor/bias	Diode + capacitor + resistor

Voltage multipliers

Diode-capacitor stages charge on alternate half cycles. Ideal light-load outputs are approximately 2Vm for a doubler and 3Vm for a tripler. Practical output is lower because of diode drops, ripple and loading.

Multiplier estimate

$V_m = 20$ V.
Ideal doubler≈ 40 V.
Ideal tripler≈ 60 V.

Expected: rectifier/filter comparison, PIV and ripple, τ and conditions, positive/negative/biased shunt clippers, slicer, clampers, doubler and tripler.

FORMULA BANK

Important formulas

Diode

$R_{DC} = V_D/I_D$

$r_{ac} = \Delta V_D/\Delta I_D$

$P_D = V_D I_D$

$I = I_S(e^{(qV_D/kT)} - 1)$

BJT

$IE = IC + IB$

$\alpha = IC/IE$; $\beta = IC/IB$; $\gamma = IE/IB$

$\beta = \alpha/(1-\alpha)$; $\alpha = \beta/(\beta+1)$; $\gamma = \beta+1$

FET and UJT

$ID = IDSS(1-VGS/VP)^2$

$\eta = RB1/(RB1+RB2)$

Wave shaping

$\tau = RC$

Half-wave $VDC = Vm/\pi$; $PIV = Vm$

Full-wave $VDC = 2Vm/\pi$; ripple=0.482

Centre-tap $PIV = 2Vm$; Bridge $PIV = Vm$

Common mistake: Convert mA to A and μF/nF to F before substitution.

SYMBOLS, CIRCUITS, CURVES AND WAVEFORMS

Drawing practice bank

Symbols

- PN and Zener diode.

- NPN and PNP.
- N-channel JFET.
- Depletion MOSFET.
- UJT, capacitor, inductor, transformer, sources and ground.

Circuits

- Diode biasing.
- CB, CE, CC and CE amplifier.
- JFET/MOSFET/UJT circuits.
- Rectifiers and filters.
- Differentiator, integrator, clipper, clamper and multipliers.

Curves & Waveforms

- Diode V-I and Zener breakdown.
- CB/CE input-output.
- JFET/MOSFET curves.
- UJT peak-valley curve.
- Rectifier, filter, RC, clipper and clamper waveforms.

EXTENDED ELECTRONICS NOTES

Useful device and circuit notes

Additional syllabus-relevant explanations developed from major electronics tutorial topics. No external tutorial links are included.

Diode family and selection

Device	Main property	Typical use	Selection point
Signal diode	Fast operation at low current	Detection, switching and small-signal clipping	Switching speed, forward current and reverse voltage
Power diode	Handles larger current and power	Rectifiers and power supplies	Average current, surge current, PIV and power loss
Zener diode	Operates safely in reverse breakdown	Voltage reference and protection	Zener voltage, current range and power rating

LED	Converts current into light	Indicators and displays	Forward voltage, current and series resistor
Schottky diode	Low forward drop and fast switching	High-frequency and low-voltage circuits	Reverse leakage and voltage rating
Bypass diode	Provides an alternate current path	Protection of series-connected cells or sections	Current and reverse-voltage capability

Practical diode choice: A rectifier diode is chosen mainly by current and PIV, a signal diode by speed and low capacitance, a Zener by breakdown voltage and power, and an LED by forward current and required light output.

Transistor switching and Darlington connection

BJT as a switch

Cut-off represents the OFF state and saturation represents the ON state. A base resistor limits base current. The load is normally placed in the collector path, and an inductive load requires a flyback diode.

> Switching circuit-□ transistor active region-□ □□□□; cut-off □□□□□□□□□□□ saturation region-□□□□ □□□□□□□□□□□□□□□□□□□□□.

Darlington pair

Two BJTs connected so the first emitter drives the second base provide very high overall current gain. The approximate gain is the product of the individual gains, but the pair has a higher effective base-emitter voltage and slower switching than a single transistor.

FET and MOSFET switches

Gate voltage controls the channel while drawing very little steady gate current. MOSFETs are preferred for efficient voltage-controlled switching; the gate must still be charged and discharged during transitions.

UJT triggering

The negative-resistance region allows a capacitor to charge slowly and discharge rapidly, producing trigger pulses. This is the basic idea of a UJT relaxation oscillator.

RC networks: charging, discharging and waveform action

Capacitor charging: $v_C(t)=V(1-e^{-t/RC})$

Capacitor discharging: $v_C(t)=V_0e^{-t/RC}$

Time	Charging level	Discharging level
1τ	About 63.2%	About 36.8%
3τ	About 95%	About 5%
5τ	Practically fully charged	Practically discharged

Wave-shaping connection: A differentiator emphasizes rapid changes and produces edge spikes. An integrator averages the input over time and changes a square wave toward a ramp or triangular waveform.

Power-supply signal path

1. Transformer

Provides isolation and the required AC level.

2. Rectifier

Changes AC into pulsating DC.

3. Filter

Reduces ripple using C, L or a combined network.

4. Regulation / Load

Maintains a useful DC level and supplies the circuit.

A capacitor-input filter gives a high no-load DC value but poorer regulation at heavy load. An inductor filter is more suitable for larger current. A π filter combines strong smoothing with practical DC output.

Clipping, clamping and multiplication

Circuit	What it does	Typical application
Positive/negative clipper	Removes one polarity beyond a set level	Amplitude limiting and protection
Biased clipper	Sets a non-zero clipping level	Threshold and slicing circuits
Combination clipper	Limits both upper and lower levels	Two-sided protection
Clamper	Shifts the complete waveform up or down	DC restoration
Voltage multiplier	Uses diode-capacitor stages for higher DC voltage	Low-current high-voltage supplies

Quick practical checks

Before power ON

- Check diode polarity and transistor pin identification.
- Confirm capacitor polarity and voltage rating.
- Verify current-limiting resistors.

During measurement

- Use a common reference ground.
- Compare input and output waveforms on the same time scale.
- Do not exceed device current, voltage or power ratings.

When output is wrong

- Check biasing first.
- Confirm the expected conduction half-cycle.
- Inspect loading, ripple and component orientation.

DETAILED TUTORIAL EXPANSION

Source-based Basic Electronics study notes

This expansion converts the relevant Diodes, Transistors, Power Electronics, RC Networks and Miscellaneous Circuits topics from the supplied tutorial index into syllabus-focused explanations. The material is paraphrased and organized for Course 2041; it is not a list of external links.

How to use this section: First read the working principle, then study the comparison table, copy the formula, and finally solve the mini example. Existing lesson material above remains the primary syllabus sequence; this section adds depth and practical connection.

1. Semiconductor material to practical diode

Intrinsic semiconductor

Pure silicon or germanium contains covalent bonds. Thermal energy can release an electron from a bond and leave a hole. The free electron and the hole form an electron-hole pair, and recombination removes the pair.

N-type material

Pentavalent donor atoms contribute an extra electron. Electrons are majority carriers and holes are minority carriers. The crystal as a whole remains electrically neutral.

P-type material

Trivalent acceptor atoms create vacancies called holes. Holes are majority carriers and electrons are minority carriers. A hole represents the movement of missing-electron charge through the lattice.

Condition	Depletion region	Barrier	Current	Useful model
Zero bias	Natural width	Built-in potential exists	Net current is approximately zero	Dynamic equilibrium
Forward bias	Narrows	Reduced	Rises rapidly after knee voltage	Closed switch with forward drop

Reverse bias	Widens	Increased	Only small leakage until breakdown	Open switch before breakdown

The depletion region contains fixed ions but very few mobile carriers. Forward bias pushes majority carriers toward the junction, while reverse bias pulls them away. A silicon diode is commonly approximated by a 0.7 V forward drop and a germanium diode by about 0.3 V, although actual values depend on current and temperature.

Signal and switching diode

Designed for low current and fast response. Important specifications are reverse-voltage rating, forward current, junction capacitance and recovery speed.

Power rectifier diode

Designed for larger forward current and surge current. Selection depends mainly on average current, peak inverse voltage, power dissipation and thermal conditions.

Zener diode

Intentionally operated in reverse breakdown. Within its permitted current range, the voltage remains nearly constant, so the device can act as a reference or simple shunt regulator.

LED

Produces light when forward biased. A series resistor is essential because a small voltage increase can otherwise cause excessive current.

LED series resistor: $R = (V_S - V_F) / I_F$

Mini example — LED resistor

Supply = 9 V, LED drop = 2 V, required current = 15 mA.
$R = (9 - 2)/0.015 = 466.7\ \Omega$.
Choose the next convenient standard value, 470 Ω.

Zener series resistor: $R_S = (V_{IN} - V_Z) / (I_L + I_Z)$

Mini example — simple Zener regulator

V_{IN}=12 V, V_Z=5.1 V, load current=10 mA, chosen Zener current=5 mA.
R_S=(12−5.1)/(0.010+0.005)=460 Ω.
Use approximately 470 Ω and verify resistor/Zener power ratings.

2. Rectification and filtering in a DC power supply

Half-wave rectifier

One diode conducts during one half-cycle and blocks the other. The ripple frequency equals the supply frequency, so smoothing is relatively difficult.

Centre-tapped full-wave

Each diode conducts on alternate half-cycles through half of the transformer secondary. Ripple frequency is twice the supply frequency, but each diode must withstand a higher PIV.

Bridge rectifier

Two diodes conduct in each half-cycle. It does not require a centre tap and gives good transformer utilization, but the conducting path contains two forward drops.

Item	Half-wave	Full-wave centre tap	Bridge
Diodes conducting at a time	1	1	2
Ripple frequency	f	2f	2f
Ideal average output	V_m/π	$2V_m/\pi$	$2V_m/\pi$
PIV per diode	V_m	$2V_m$	V_m
Main advantage	Simplicity	One diode drop in path	No centre tap, better utilization

A capacitor-input filter charges near each rectified peak and supplies the load between peaks. Increasing capacitance or decreasing load current reduces ripple. A series inductor opposes changes in current, and a π filter combines shunt capacitors with a series element for stronger smoothing.

Approximate capacitor-filter ripple: $\Delta V \approx I_L / (f_{ripple} C)$

Mini example — ripple estimate

Full-wave rectifier on 50 Hz supply: f_{ripple}=100 Hz.
I_L=50 mA and C=1000 μF.
ΔV≈0.05/(100×0.001)=0.5 V peak-to-peak (approximate).

3. BJT operation, configurations and switching

A bipolar junction transistor contains two PN junctions and three terminals: emitter, base and collector. In normal active operation the emitter-base junction is forward biased and the collector-base junction is reverse biased. A small base current controls a much larger collector current.

Region	Junction condition	Collector current	Application
Cut-off	Base-emitter not sufficiently forward biased	Approximately zero	Switch OFF
Active	EB forward, CB reverse	$I_C \approx \beta I_B$	Amplification
Saturation	Both junctions forward biased	Limited mainly by load circuit	Switch ON

Common base

Low input resistance, high output resistance, current gain below unity, useful voltage gain and good high-frequency response.

Common emitter

Useful current and voltage gain, high power gain and 180° phase reversal. It is the most common general amplifier configuration.

Common collector

Voltage gain near unity, high input resistance and low output resistance. It is widely used as an emitter follower or buffer.

$I_E = I_C + I_B$; $\beta = I_C / I_B$; $\alpha = I_C / I_E$

Mini example — saturated transistor switch

A 5 V logic signal drives an NPN transistor. Required collector current is 60 mA.
Use forced $\beta=10$ for reliable saturation, so $I_B=60/10=6$ mA.
With $V_{BE}\approx0.7$ V, $R_B=(5-0.7)/0.006\approx717\ \Omega$.
Choose about 680 Ω or 750 Ω after checking source-current capability.

Inductive-load warning: A relay coil or motor stores magnetic energy. When transistor current is interrupted, the coil can generate a large reverse voltage. Connect a flyback diode across a DC coil with polarity that is reverse biased during normal operation.

4. JFET and MOSFET control of current

JFET

The gate-channel junction is normally reverse biased. Changing V_{GS} changes depletion width and therefore the effective channel width. Gate current is extremely small, giving high input impedance.

For an N-channel JFET, making V_{GS} more negative reduces I_D. At $V_{GS(off)}$ the channel is effectively cut off.

Depletion MOSFET

An insulated gate controls an existing channel. It can conduct at $V_{GS}=0$; one gate polarity depletes the channel and the opposite polarity enhances it.

The oxide insulation produces extremely high DC input resistance, but gate capacitance must still be charged and discharged during switching.

JFET transfer approximation: $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$

Mini example — JFET drain current

I_{DSS}=12 mA, V_P=−6 V, V_{GS}=−3 V.
$I_D=12(1-(-3/-6))^2=12(0.5)^2=3$ mA.

Feature	BJT	JFET	MOSFET
Control quantity	Base current	Gate voltage	Gate voltage
Carrier type	Bipolar	Unipolar	Unipolar
Input impedance	Lower	High	Very high
Typical strength	Current gain and simple analogue stages	Low gate current and low-noise input	Efficient voltage-controlled switching

5. UJT negative resistance and relaxation oscillator

A unijunction transistor has one PN junction and three terminals: emitter E, base B1 and base B2. The N-type bar between B1 and B2 behaves like two series resistances. The emitter junction is connected to their internal division point.

$$R_{BB}=R_{B1}+R_{B2};\quad \eta=R_{B1}/R_{BB};\quad V_P\approx\eta V_{BB}+V_D$$

Below the peak voltage, the emitter junction is not conducting. When the emitter reaches the peak point, carrier injection reduces the effective resistance toward B1. Emitter current increases while emitter voltage falls, producing the negative-resistance region. Conduction continues toward the valley point, after which the UJT returns to the OFF state when current becomes too small.

Charging interval

The timing capacitor charges through a resistor. Its voltage rises exponentially toward the supply while the UJT remains OFF.

Trigger and discharge

When capacitor voltage reaches the UJT peak voltage, the emitter switches ON and the capacitor discharges quickly through B1, producing a narrow trigger pulse.

Mini example — UJT peak voltage

$\eta=0.65$, $V_{BB}=10$ V, silicon emitter junction drop≈0.7 V.
$V_P \approx 0.65\times10+0.7=7.2$ V.

6. RC charging, discharging and pulse response

$$\text{Charging: } v_C(t)=V(1-e^{-t/RC})$$

$$\text{Discharging: } v_C(t)=V_0e^{-t/RC}$$

Elapsed time	Charging voltage	Discharging voltage
1τ	63.2% of final	36.8% remains
2τ	86.5%	13.5% remains
3τ	95.0%	5.0% remains
5τ	About 99.3%	About 0.7% remains

Mini example — capacitor charging

$V=10$ V, $R=100$ kΩ, $C=10$ μF, so $\tau=1$ s.
At $t=1$ s: $v_C=10(1-e^{-1})\approx6.32$ V.
At $t=3$ s: $v_C\approx9.50$ V.

7. RC integrator and differentiator

RC integrator

Input is applied through the resistor and output is taken across the capacitor. The capacitor cannot follow rapid changes fully, so a square wave can become a ramp-like or approximately triangular waveform.

$$\text{Good integration: RC much greater than pulse width (or } X_C \ll R \text{ at the signal frequency)}$$

RC differentiator

Input is applied through the capacitor and output is taken across the resistor. Rapid changes pass strongly, so square-wave edges produce a positive spike at the rising edge and a negative spike at the falling edge.

Good differentiation: RC much smaller than pulse width (or $X_C \gg R$ at the signal frequency)

Memory rule: Output across C tends toward integration; output across R tends toward differentiation. Always compare RC with pulse width or period before predicting the waveform.

8. Clippers, clampers and voltage multipliers

Circuit	Main action	What changes	Typical purpose
Series/shunt clipper	Blocks or diverts a selected part	Waveform amplitude/shape	Limiting and protection
Biased clipper	Sets a non-zero clipping threshold	Upper or lower limit	Slicing and threshold control
Combination clipper	Limits both polarities	Upper and lower boundaries	Two-sided protection
Clamper	Adds a DC level to the waveform	Reference level; ideal peak-to-peak value stays similar	DC restoration and level shifting
Voltage multiplier	Charges capacitors on alternate half-cycles	Higher DC voltage at light load	Low-current high-voltage supply

In a clamper, the diode establishes the charging polarity and the capacitor stores a voltage that shifts the entire waveform. For effective clamping, the discharge time constant through the load should be long compared with the signal period. In a voltage doubler or tripler, loading, diode drops and ripple make the practical output lower than the simple ideal multiples.

Mini example — clipping level

A silicon biased clipper has a 3 V bias source in series with the diode.
A first estimate of the clipping threshold is 3 V + 0.7 V = 3.7 V.
The actual polarity and whether positive or negative peaks are limited depend on diode orientation and bias-source polarity.

9. Source-topic coverage map

Uploaded tutorial category	Course 2041 use	Topics converted into notes
Semiconductor Diodes	Module 1 and Module 4	PN junction, bias, rectification, Zener, LED, clipping and clamping
Transistors	Module 2 and Module 3	BJT regions/configurations, transistor switch, JFET and MOSFET
Power Electronics	Module 3	UJT construction, negative resistance, triggering and relaxation oscillator
RC Networks	Module 4	Charging, discharging, time constant, waveforms, integration and differentiation
Miscellaneous Circuits / Power Supplies	Module 4	Voltage multiplication, rectifier-filter chain and practical waveform control

10. Quick concept test

1. Why does forward bias reduce depletion width?
2. Why is a resistor required with an LED?
3. Why does full-wave rectification simplify filtering?
4. Which BJT region is used for linear amplification?
5. Why does a MOSFET have very high input resistance?

6. What produces the negative-resistance region of a UJT?
7. What percentage of final voltage is reached after one RC time constant?
8. Where is the output taken in an RC differentiator?
9. How does a clamper differ from a clipper?
10. Why is practical multiplier output below the ideal value?

Answer outline: majority carriers are pushed toward the junction; current must be limited; ripple frequency doubles; active region; insulated gate; emitter injection lowers the B1-side resistance; 63.2%; across R; clamper shifts while clipper removes/limits; diode drops, ripple and loading reduce output.

EXPECTED QUESTIONS

Module-wise question bank

Module 1

1. Energy-band diagrams.
2. Intrinsic/extrinsic and doping.
3. PN-junction formation.
4. Diode V-I and specifications.
5. Resistance and power numericals.

Module 2

1. NPN/PNP construction.
2. Modes and currents.
3. Gain relations.
4. Configuration comparison.
5. CE characteristics and amplifier.

Module 3

1. JFET pinch-off.
2. MOSFET depletion operation.
3. UJT negative resistance.
4. Device comparisons.
5. η numerical.

Module 4

1. Rectifier/filter comparison.
2. Differentiator/integrator.
3. Clipper and clamper waveforms.
4. Voltage multipliers.

PRACTICE

Problems and self-assessment

Module 1

1. $V_D=0.6$ V, $I_D=20$ mA. **$R_{DC}=30\ \Omega$**
2. $\Delta V=0.03$ V, $\Delta I=5$ mA. **$r_{ac}=6\ \Omega$**

Module 2

1. $I_C=9.9$ mA, $I_B=0.1$ mA. **$I_E=10$ mA, $\beta=99$**
2. $\alpha=0.975$. **$\beta=39$**

Module 3

1. $R_{B1}=7\ k\Omega, R_{B2}=3\ k\Omega$. **$\eta=0.7$**
2. $I_{DSS}=8$ mA, $V_P=-3$ V, $V_{GS}=-1.5$ V. **$I_D=2$ mA**

Module 4

1. $R=20\ k\Omega, C=0.005\ \mu F$. **$\tau=0.1$ ms**
2. $V_m=25$ V. **Doubler≈50 V**

Checklist

- ✓ I can draw all devices.
- ✓ I can read curves.
- ✓ I can solve formulas.
- ✓ I can predict waveforms.

Fill: BJT base is ____ and lightly doped. Full-wave ripple≈____. UJT has ____ junction. **Answers:** thin, 0.482, one.

QUICK REVISION

Last-day revision

M1

- Donor → N; acceptor → P.
- Forward bias narrows depletion.
- RDC=VD/ID.

M2

- Active amplifies.
- Cutoff/saturation switch.
- CE high power gain.

M3

- FET voltage controlled.
- MOS gate insulated.
- UJT negative resistance.

M4

- Bridge PIV=Vm.
- τ=RC.
- Clipper cuts; clamper shifts.

Common Examination Mistakes: wrong transistor arrow, missing axes on curves, mixing α and β, forgetting ideal-diode assumptions, drawing series clippers when shunt type is asked, writing RC≫T for differentiator, writing RC≪T for integrator, and not converting mA/μA/nF/μF into base units.

MODEL QUESTION PAPER

BASIC ELECTRONICS • 75 Marks • 3 Hours

PART A — 10 × 1 = 10

1. Define forbidden gap.
2. Name majority carrier in P-type.
3. Write diode static-resistance formula.
4. State BJT current relation.
5. Which BJT mode amplifies?
6. Why is FET voltage-controlled?
7. Define η.
8. State ideal full-wave ripple factor.
9. State integration condition.
10. What does a clamper change?

PART B — Answer any five, 5 × 5 = 25

1. Compare conductor, semiconductor and insulator.
2. VD=0.72 V, ID=12 mA: find resistance and power.
3. Explain BJT modes.
4. IC=8 mA, β=80: find IB and IE.
5. Compare BJT and JFET.
6. Explain capacitor, inductor and π filters.
7. R=50 kΩ,C=0.02 μF: find τ and check differentiation for T=10 ms.

PART C — Answer any four, 4 × 10 = 40

1. Explain intrinsic/extrinsic semiconductors and junction formation.
2. Explain diode V-I characteristic and resistances.
3. Explain NPN operation, gains, configurations and CE amplifier.
4. Explain JFET and depletion MOSFET.
5. Explain UJT construction and characteristic.
6. Compare rectifiers and explain filters.
7. Explain RC differentiator and integrator.
8. Explain clippers, clampers and multipliers.

COMPLETE MODEL ANSWERS

Answer key and full solutions

Part A

1	Energy range between valence and conduction bands with no allowed states.
2	Holes.
3	RDC=VD/ID.
4	IE=IC+IB.
5	Active mode.
6	Drain current is controlled by gate voltage with negligible gate current.
7	η=RB1/(RB1+RB2).
8	0.482.
9	RC≫T.
10	DC level.

Part B

B1

Conductor bands overlap; semiconductor has a small gap; insulator has a large gap. Draw and label VB, CB and Eg.

B2

ID=0.012 A. RDC=0.72/0.012=60 Ω. PD=0.72×0.012=0.00864 W=8.64 mW.

B3

Cutoff: OFF. Active: EB forward and CB reverse, amplification. Saturation: both junctions forward, ON.

B4

$I_B = I_C/\beta = 8/80 = 0.1$ mA. $I_E = 8 + 0.1 = 8.1$ mA.

B5

BJT is bipolar/current-controlled with lower input impedance. JFET is unipolar/voltage-controlled with high input impedance and reverse-biased gate.

B6

Capacitor fills waveform valleys; inductor blocks ripple current; π filter uses C-L-C for strong smoothing.

B7

$\tau = 50{,}000 \times 0.02 \times 10^{-6} = 0.001$ s $= 1$ ms. Since τ is smaller than 10 ms, approximate differentiation is possible.

Part C

C1

Intrinsic semiconductor is pure; thermal generation creates equal electrons and holes. Donor doping creates N-type; acceptor doping creates P-type. After joining, carrier diffusion and recombination leave fixed ions and form depletion region and barrier potential. Drift balances diffusion at equilibrium.

C2

Forward current rises rapidly after knee voltage. Reverse leakage remains small until breakdown. Label axes, knee, leakage and breakdown. $R_{DC}=V_D/I_D$; $r_{ac}=\Delta V_D/\Delta I_D$.

C3

NPN active operation injects electrons from emitter; small recombination gives I_B and most carriers form I_C. $I_E=I_C+I_B$; $\alpha=I_C/I_E$; $\beta=I_C/I_B$. CE provides voltage and current gain with 180° phase reversal.

C4

JFET reverse gate bias changes depletion width and channel current. Pinch-off gives nearly constant I_D. Depletion MOSFET conducts at $V_{GS}=0$; negative V_{GS} reduces current and positive V_{GS} enhances it.

C5

UJT has E,B1,B2 and equivalent R_{B1}/R_{B2}. At peak point emitter conduction reduces R_{B1}, creating negative resistance until valley point. $\eta=R_{B1}/(R_{B1}+R_{B2})$.

C6

Half-wave: $V_{DC}=V_m/\pi$, ripple=1.21, PIV=V_m, TUF≈0.287. Full-wave: $V_{DC}=2V_m/\pi$, ripple=0.482. Centre tap PIV=$2V_m$; bridge PIV=V_m and TUF≈0.812. Filters reduce ripple using C, L or C-L-C.

C7

Differentiator uses series C/shunt R and $RC \ll T$; square input gives spikes. Integrator uses series R/shunt C and $RC \gg T$; square input gives triangular output.

C8

Shunt clippers remove selected positive/negative portions; bias sets level. Clampers shift DC level using diode-capacitor-resistor. Doublers and triplers produce approximately 2Vm and 3Vm under ideal light-load conditions.

REFERENCES

Books and online resources

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R1	N. N. Bhargava, Kulshreshtha and S. C. Gupta, Basic Electronics and Linear Circuits, TMH.
R2	Robert Boylestad, Electronic Devices and Circuits, PHI.
R3	Anil K. Maini and Varsha Agarwal, Electronic Devices and Circuits, Wiley India.
R4	David A. Bell, Electronic Devices and Circuits, PHI.

Electronics Tutorials

ElProCus

BrainKart

Electrical4U

TutorialsPoint

FINAL EXAMINATION CHECKLIST

Before the exam

- ✓ I can draw structures and symbols.
- ✓ I can label characteristic regions.
- ✓ I can solve diode/BJT/FET/UJT formulas.

- ✓ I can compare rectifiers and filters.
- ✓ I can calculate τ.
- ✓ I can draw clipper and clamper waveforms.

Best answer order: definition → construction/symbol → bias → working → characteristic/waveform → formula → application.